

Rules of Equivalence

in Propositional Logic

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- ① What is Equivalence?
- ② Rules
- ③ Applications

Rules of Equivalence

1. DeMorgan's Rule (DM)

$$\neg(p \wedge q) \Leftrightarrow (\neg p \vee \neg q)$$

$$\neg(p \vee q) \Leftrightarrow (\neg p \wedge \neg q)$$

Proof

p	q	$\neg p$	$\neg q$	$(p \wedge q)$	$\neg(p \wedge q)$	$\stackrel{?}{\Leftrightarrow}$	$(\neg p \vee \neg q)$
F	F	T	T	F	.	$\stackrel{?}{=}$	.
F	T	T	F	F	.	$\stackrel{?}{=}$	.
T	F	F	T	F	.	$\stackrel{?}{=}$	.
T	T	F	F	T	.	$\stackrel{?}{=}$	.

Rules of Equivalence

1. DeMorgan's Rule (DM)

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Proof

p	q	$\neg p$	$\neg q$	$(p \wedge q)$	$\neg(p \wedge q)$	$\Leftrightarrow$	$(\neg p \vee \neg q)$
F	F	T	T	F	T	=	T
F	T	T	F	F	T	=	T
T	F	F	T	F	T	=	T
T	T	F	F	T	F	=	F

Rules of Equivalence

2. Commutativity (Com)

$$(p \vee q) \Leftrightarrow (q \vee p)$$

$$(p \wedge q) \Leftrightarrow (q \wedge p)$$

Proof

p	q	$(\textcolor{red}{p} \wedge \textcolor{blue}{q})$	$\overset{?}{\Leftrightarrow}$	$(\textcolor{blue}{q} \wedge \textcolor{red}{p})$
F	F	.	$\overset{?}{=}$	.
F	T	.	$\overset{?}{=}$	.
T	F	.	$\overset{?}{=}$	.
T	T	.	$\overset{?}{=}$	.

2. Commutativity (Com)

$$(p \vee q) \Leftrightarrow (q \vee p)$$

$$(p \wedge q) \Leftrightarrow (q \wedge p)$$

Proof

p	q	$(p \wedge q)$	$\Leftrightarrow$	$(q \wedge p)$
F	F	F	=	F
F	T	F	=	F
T	F	F	=	F
T	T	T	=	T

Rules of Equivalence

3. Associativity (Assoc)

$$[p \vee (q \vee r)] \Leftrightarrow [(p \vee q) \vee r]$$

$$[p \wedge (q \wedge r)] \Leftrightarrow [(p \wedge q) \wedge r]$$

Proof

p	q	r	$(q \vee r)$	$(p \vee q)$	$[p \vee (q \vee r)]$	$\Leftrightarrow$	$[(p \vee q) \vee r]$
F	F	F	F	F	.	$\frac{?}{=}$	.
F	F	T	T	F	.	?	.
F	T	F	T	T	.	?	.
F	T	T	T	T	.	?	.
T	F	F	F	T	.	?	.
T	F	T	T	T	.	?	.
T	T	F	T	T	.	?	.
T	T	T	T	T	.	?	.

Rules of Equivalence

3. Associativity (Assoc)

$$[p \vee (q \vee r)] \Leftrightarrow [(p \vee q) \vee r]$$

$$[p \wedge (q \wedge r)] \Leftrightarrow [(p \wedge q) \wedge r]$$

Proof

p	q	r	$(q \vee r)$	$(p \vee q)$	$[p \vee (q \vee r)]$	$\Leftrightarrow$	$[(p \vee q) \vee r]$
F	F	F	F	F	F	=	F
F	F	T	T	F	T	=	T
F	T	F	T	T	T	=	T
F	T	T	T	T	T	=	T
T	F	F	F	T	T	=	T
T	F	T	T	T	T	=	T
T	T	F	T	T	T	=	T
T	T	T	T	T	T	=	T

Rules of Equivalence

4. Distribution (Dist)

$$[p \wedge (q \vee r)] \Leftrightarrow [(p \wedge q) \vee (p \wedge r)]$$

$$[p \vee (q \wedge r)] \Leftrightarrow [(p \vee q) \wedge (p \vee r)]$$

Proof. $s \triangleq [p \wedge (q \vee r)]$; $t \triangleq [(p \wedge q) \vee (p \wedge r)]$

p	q	r	$(q \vee r)$	$(p \wedge q)$	$(p \wedge r)$	s	$\overset{?}{\Leftrightarrow}$	t
F	F	F	F	F	F	.	$\overset{?}{=}$	.
F	F	T	T	F	F	.	?	.
F	T	F	T	F	F	.	?	.
F	T	T	T	F	F	.	?	.
T	F	F	F	F	F	.	?	.
T	F	T	T	F	T	.	?	.
T	T	F	T	T	F	.	?	.
T	T	T	T	T	T	.	?	.

Rules of Equivalence

4. Distribution (Dist)

$$[p \wedge (q \vee r)] \Leftrightarrow [(p \wedge q) \vee (p \wedge r)]$$

$$[p \vee (q \wedge r)] \Leftrightarrow [(p \vee q) \wedge (p \vee r)]$$

Proof. $s \triangleq [p \wedge (q \vee r)]$; $t \triangleq [(p \wedge q) \vee (p \wedge r)]$

p	q	r	$(q \vee r)$	$(p \wedge q)$	$(p \wedge r)$	s	$\Leftrightarrow$	t
F	F	F	F	F	F	F	=	F
F	F	T	T	F	F	F	=	F
F	T	F	T	F	F	F	=	F
F	T	T	T	F	F	F	=	F
T	F	F	F	F	F	F	=	F
T	F	T	T	F	T	T	=	F
T	T	F	T	T	F	T	=	F
T	T	T	T	T	T	T	=	F

Rules of Equivalence

5. Double negation (DN)

$$p \Leftrightarrow \neg\neg p$$

Proof

p	$\neg p$	p	$\overset{?}{\Leftrightarrow}$	$\neg\neg p$
F	T	.	$\overset{?}{=}$	.
T	F	.	$\overset{?}{=}$	.

5. Double negation (DN)

$$p \Leftrightarrow \neg\neg p$$

Proof

p	$\neg p$	p	$\Leftrightarrow$	$\neg\neg p$
F	T	F	=	F
T	F	T	=	T

Rules of Equivalence

6. Transposition (Trans)

$$(p \Rightarrow q) \Leftrightarrow (\neg q \Rightarrow \neg p)$$

Proof

p	q	$\neg q$	$\neg p$	$(p \Rightarrow q)$	$\overset{?}{\Leftrightarrow}$	$(\neg q \Rightarrow \neg p)$
F	F	T	T	.	$\overset{?}{=}$	.
F	T	F	T	.	$\overset{?}{=}$	.
T	F	T	F	.	$\overset{?}{=}$	.
T	T	F	F	.	$\overset{?}{=}$	.

6. Transposition (Trans)

$$(p \Rightarrow q) \Leftrightarrow (\neg q \Rightarrow \neg p)$$

Proof

p	q	$\neg q$	$\neg p$	$(p \Rightarrow q)$	$\Leftrightarrow$	$(\neg q \Rightarrow \neg p)$
F	F	T	T	T	=	T
F	T	F	T	T	=	T
T	F	T	F	F	=	F
T	T	F	F	T	=	T

7. Material implication (Impl)

$$(p \Rightarrow q) \Leftrightarrow (\neg p \vee q)$$

Proof

p	q	$\neg p$	$(p \Rightarrow q)$	$\stackrel{?}{\Leftrightarrow}$	$(\neg p \vee q)$
F	F	T	.	$\stackrel{?}{=}$	.
F	T	T	.	$\stackrel{?}{=}$	.
T	F	F	.	$\stackrel{?}{=}$	.
T	T	F	.	$\stackrel{?}{=}$	.

7. Material implication (Impl)

$$(p \Rightarrow q) \Leftrightarrow (\neg p \vee q)$$

Proof

p	q	$\neg p$	$(p \Rightarrow q)$	$\Leftrightarrow$	$(\neg p \vee q)$
F	F	T	T	=	T
F	T	T	T	=	T
T	F	F	F	=	F
T	T	F	T	=	T

Rules of Equivalence

8. Material equivalence (Equiv)

$$[p \Leftrightarrow q] \Leftrightarrow [(p \Rightarrow q) \wedge (q \Rightarrow p)]$$

$$[p \Leftrightarrow q] \Leftrightarrow [(p \wedge q) \vee (\neg p \wedge \neg q)]$$

Proof

p	q	$(p \Rightarrow q)$	$(q \Rightarrow p)$	$[p \Leftrightarrow q]$	$\stackrel{?}{\Leftrightarrow}$	$[(p \Rightarrow q) \wedge (q \Rightarrow p)]$
F	F	T	T	.	$\stackrel{?}{=}$	.
F	T	T	F	.	$\stackrel{?}{=}$	.
T	F	F	T	.	$\stackrel{?}{=}$	.
T	T	T	T	.	$\stackrel{?}{=}$	.

8. Material equivalence (Equiv)

$$[p \Leftrightarrow q] \Leftrightarrow [(p \Rightarrow q) \wedge (q \Rightarrow p)]$$

$$[p \Leftrightarrow q] \Leftrightarrow [(p \wedge q) \vee (\neg p \wedge \neg q)]$$

Proof

p	q	$(p \Rightarrow q)$	$(q \Rightarrow p)$	$[p \Leftrightarrow q]$	$\Leftrightarrow$	$[(p \Rightarrow q) \wedge (q \Rightarrow p)]$
F	F	T	T	T	=	T
F	T	T	F	F	=	F
T	F	F	T	F	=	F
T	T	T	T	T	=	T

Rules of Equivalence

9. Exportation (Exp))

$$[(p \wedge q) \Rightarrow r] \Leftrightarrow [p \Rightarrow (q \Rightarrow r)]$$

Proof

p	q	r	$(p \wedge q)$	$(q \Rightarrow r)$	$[(p \wedge q) \Rightarrow r]$	$\Leftrightarrow$	$[p \Rightarrow (q \Rightarrow r)]$
F	F	F	F	T	.	$\frac{?}{=}$	.
F	F	T	F	T	.	?	.
F	T	F	F	F	.	?	.
F	T	T	F	T	.	?	.
T	F	F	F	T	.	?	.
T	F	T	F	T	.	?	.
T	T	F	T	F	.	?	.
T	T	T	T	T	.	?	.

Rules of Equivalence

9. Exportation (Exp))

$$[(p \wedge q) \Rightarrow r] \Leftrightarrow [p \Rightarrow (q \Rightarrow r)]$$

Proof

p	q	r	$(p \wedge q)$	$(q \Rightarrow r)$	$[(p \wedge q) \Rightarrow r]$	$\stackrel{?}{\Leftrightarrow}$	$[p \Rightarrow (q \Rightarrow r)]$
F	F	F	F	T	T	=	T
F	F	T	F	T	T	=	T
F	T	F	F	F	T	=	T
F	T	T	F	T	T	=	T
T	F	F	F	T	T	=	T
T	F	T	F	T	T	=	T
T	T	F	T	F	F	=	F
T	T	T	T	T	T	=	T

10. Tautology (Taut)

$$p \Leftrightarrow (p \vee p)$$

$$p \Leftrightarrow (p \wedge p)$$

Proof

p	p	$\overset{?}{\Leftrightarrow}$	$(p \vee p)$
F	.	$\underline{=}$	.
T	.	$\underline{=}$	.

p	p	$\overset{?}{\Leftrightarrow}$	$(p \wedge p)$
F	.	$\underline{=}$	.
T	.	$\underline{=}$	.

10. Tautology (Taut)

$$p \Leftrightarrow (p \vee p)$$

$$p \Leftrightarrow (p \wedge p)$$

Proof

p	$p \Leftrightarrow (p \vee p)$
F	F = F
T	T = T

p	$p \Leftrightarrow (p \wedge p)$
F	F = F
T	T = T